Université Paris Cité

Quantum Devices Master Program

Internship report 2023

# Variational learning of quantum states

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**Lab presentation**

Centre de Physique Théorique (CPHT) is a renowned research institution and center of excellence for theoretical physics. The CPHT is affiliated with École Polytechnique and CNRS (Centre National de la Recherche Scientifique). It conducts cutting-edge research in various areas of theoretical physics, including particle physics, condensed matter physics, statistical physics, and quantum field theory. The CPHT collaborates with national and international institutions and plays a significant role in advancing theoretical physics research in France.

## Abstract

*The objective of Quantum State Tomography (QST) is to reconstruct the wavefunction or density matrix of an unknown quantum state based on measurement data. QST plays a crucial role in assessing the quality of quantum devices. However, fully reconstructing a many-body quantum state requires an exponentially increasing number of measurements with the system's size. In this report, we leverage the power of Machine Learning (ML) techniques to approximate the target state using neural quantum states, specifically the Restricted Boltzmann Machine (RBM). We develop a ML algorithm and evaluate its performance by learning the highly-correlated ground state of the 1D Transverse Field Ising model (TFIM) using synthetic data. We analyze the accuracy of the reconstructed state in relation to the dataset size and observe an exponential dependence on the system's size, while inversely proportional to the dataset size. Furthermore, we assess the ML algorithm's performance using more realistic synthetic data generated from a thermal state and quantum depolarization noise model. Our findings highlight the potential of ML algorithms, particularly neural quantum states, in approximating and reconstructing many-body quantum states, offering a promising avenue for Quantum State Tomography.*

# Contents

# 1 Introduction

In various scientific domains, complex problems arise where explicit algorithms cannot reliably provide accurate solutions for all possible input configurations. However, recent advancements have demonstrated that by amassing extensive datasets which describe the expected behavior of an ideal algorithm, it becomes feasible to employ "machine learning" meta-algorithms capable of generating programs to tackle the original task effectively.

For instance, the application of machine learning algorithms to analyze historical climate data enables the prediction of future climate patterns. By training models using data encompassing temperature, precipitation, and atmospheric variables, machine learning aids scientists in making more precise forecasts concerning climate change, extreme weather phenomena, and other environmental factors [1].

Physicists both benefit from and contribute to the field of machine learning (ML), as many fundamental concepts in ML draw inspiration from physics. Energy-based models in ML, for example, are influenced by the physics concept of energy, enabling the modeling of complex dependencies and interactions among variables through joint probability distributions [2]. ML has found early applications in fields like experimental High Energy Physics, which handle large datasets from particle collisions and employ ML for classification tasks [3]. Moreover, the potential applications of ML extend beyond these domains, encompassing the realm of quantum physics.

With the advent of quantum computers, the intersection between quantum computing and ML has emerged as a significant field of research. Quantum Machine Learning aims to leverage quantum systems or algorithms to enhance computational capabilities and improve the efficiency of solving machine learning tasks [4]. Additionally, ML techniques are employed in the development and study of quantum computers. Quantum control involves the application of external fields or manipulations to guide and control the dynamics of quantum systems, achieving specific objectives like coherent state transfer and optimized quantum operations [5]. ML algorithms are utilized in this context to determine optimal configurations of quantum gates, minimizing decoherence effects.

Another important algorithm in the field of quantum information is Quantum State Tomography (QST) which addresses the challenge of reconstructing the quantum state (wave function or density matrix) of an unknown quantum system using experimentally available measurements. QST plays a central role in the field of quantum technologies, serving as a valuable tool for assessing the quality and limitations of experimental platforms. Moreover, it enables the efficient extraction of information about the resulting quantum state and facilitates the simultaneous computation of multiple observables or those that are otherwise challenging to measure directly [6]. By employing QST, researchers gain deeper insights into the characteristics of quantum systems, aiding in the advancement and optimization of various quantum technologies.

However, reconstructing the state typically requires an exponential number of measurements due to the increasing number of parameters needed to fully characterize the quantum state as the system size grows. This presents challenges in obtaining precise and complete information about the quantum state with a feasible number of measurements. To address this issue, several strategies have been proposed. Tensor Networks (TN) provide an efficient encoding treating quantum states with a smaller computational cost. TN describes the wave function of the system using a network of interconnected sets of contracted tensors. A particular type of TN, Matrix Product States (MPS), is particularly effective in representing low-entangled states [7]. However, for highly entangled quantum states resulting from deep quantum circuits or high-dimensional physical systems, MPS performs poorly [8].

Neural network quantum states provide a flexible and scalable alternative to MPS for representing highly entangled quantum systems, offering the ability to capture complex correlations and optimize parameters using ML techniques. In this report, we discuss the application of ML algorithms for reconstructing the wave function from synthetic observable measurements. In Section 1, we briefly introduce ML, including the algorithms used for learning, and discuss different artificial Neural network architectures that can be employed to represent many-body quantum states. Although it is distinct from QST, we explore Variational Monte Carlo in Section 2 to investigate its capabilities in finding the ground state of the 1D Transverse Field Ising model. We use this state as a "toy model" to evaluate the performance of the developed quantum tomography algorithm. Section 3 focuses on Quantum State Reconstruction (QSR), explaining its principles and showcasing the use of Kullback-Leibler divergence to extract the wave function from synthetic data measurements. Additionally, we present a preliminary result on reconstructing the Bell states. Finally, in the last section, we apply the developed ML algorithm to reconstruct the ground state of the 1D Transverse Field Ising model. We investigate its limitations and evaluate its performance on synthetic data generated by the thermalization model and quantum depolarization noise model.

## 1.1 A brief introduction to Machine learning

Machine learning (ML) is a branch of artificial intelligence that studies algorithms and statistical models that computers use to perform tasks without explicit instructions. ML finds applications in diverse fields, such as image and speech recognition, recommendation systems, and autonomous vehicles, continually advancing our ability to leverage data for intelligent problem-solving. Through the analysis of patterns and relationships within the data, ML algorithms can make predictions, classify objects, recognize patterns, and reveal hidden insights. This process involves training models using labeled examples to generalize and accurately predict or decide upon unseen data.

ML encompasses different methodologies that are applied depending on the nature of the problem at hand. One widely used approach is supervised learning, which forms the focus of this report. In supervised learning, a model is trained using labeled examples, where each example consists of input features and corresponding target values. These labeled examples serve as a guide for the model to make predictions or classify new, unseen data. Mathematically, a dataset in supervised learning can be represented as $\mathcal{D} = \{(x_i, y_i = F(x_i)), i = [1, .., N]\}$, where $x_i$ represents the input features and $y_i$ represents the corresponding target values generated by an unknown target function $F(x)$. The objective is to construct a computationally cheap function $f(x)$ that approximates $F(x)$. We parametrize the space of those computationally cheap functions with a variational function $f(x; \theta)$ where $\theta$ are the variational parameters. The parameters $\theta$ are determined by optimizing a cost function (or loss function) $\mathcal{C}(y, f(x; \theta))$, which quantifies some distance between the model's predicted outputs and the true outputs. To determine the optimal parameters $\hat{\theta}$ for our model, we seek to minimize the cost function, yielding:

$$\hat{\theta} = \min_{\theta} \{\mathcal{C}(y, f(x;\theta))\} . \tag{1.1}$$

For high-dimensional optimization problems involving a large number of parameters, local, gradient-based optimization schemes are often the most viable approach. Global optimization algorithms, while effective for certain scenarios, may face computational and scalability challenges when dealing with several thousand to billions of parameters. In such cases, local optimization methods leverage the local information provided by gradients to iteratively update parameter values and search for optimal solutions. By navigating the parameter space based on the direction of steepest descent, these algorithms can efficiently converge towards a local minimum.

A prototypical, albeit simple, ML algorithm is linear regression. Assuming that $x$ is a scalar, the linear model is $f(x; a, b) = ax + b$ and the objective is to find the value of the real parameters $a$ and $b$, such that the cost function,

$$\mathcal{C}(y, f(x;\theta)) = \sum_{i=1}^{N} (y_i - f(x_i;\theta))^2, \tag{1.2}$$

which is the squared error is minimized.

Having introduced the concepts of ML, we now turn our attention to the optimization methods employed to find the optimal parameters. Given that the number of parameters in our model is on the order of $10^2$, utilizing global optimization methods would entail significant computational costs. Therefore, we opt for local optimization techniques, specifically gradient descent algorithms. These iterative approaches enable us to update the model's parameters based on the gradients of the cost function, gradually converging toward an optimal solution.

### 1.1.1 Gradient descent method

The parameters which minimize the cost function are obtained through learning algorithms of which the most powerful and widely used classes are gradient descent and its generalizations. These algorithms leverage the gradient of the cost function (or some higher order derivative), $E(\theta) = \mathcal{C}(y, f(x;\theta))$, to iteratively update the parameters. We say that we converged to the result when the gradient of the parameters, $\nabla_\theta E(\theta)$, approaches zero, such that doing more iterations does not reduce notably the value of the loss function.

The most basic learning algorithm is the vanilla gradient descent algorithm (GD) which is shown in Algorithm 1. The learning rate $\eta$ determines the speed at which the algorithm explores the parameter space. If the learning rate is too small, convergence may be achieved, but at the expense of longer computation time. Conversely, if the learning rate is too large, the algorithm may fail to reach the minimum and instead oscillate around it. Striking a balance by choosing an appropriate value for $\eta$ is vital for effective convergence in ML algorithms. Selecting the ideal learning rate is an important problem in ML. The ideal rate would be determined by higher-order derivatives such as the Hessian, but as the cost to compute those is very high, the rate is often determined through more empirical approximations.

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**Algorithm 1** Gradient descent algorithm

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**Require:** $\theta_0$: Initial parameters
**Require:** $\eta$ : Learning rate
**Require:** $E(\theta)$ : Cost function
  **while** $\theta_t$ do not converge **do**
    $v_t \leftarrow \eta \cdot \nabla_\theta E(\theta_t)$
    $\theta_{t+1} \leftarrow \theta_t - v_t$
  **end while**

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The simplicity of Vanilla GD determines its limitations. Being deterministic, it might get stuck on local minima. Because in ML we are often dealing with extremely rugged landscapes with several local minima, GD may get stuck close to these points. Second, gradients are computationally expensive to compute over the entirety of large data sets. For instance, the squared-error (1.2) is a sum over the whole data set. Repeating this computation at each iteration would be extremely computationally expensive, leading to poor performance.

To address these challenges, a standard approach known as minibatching divides the dataset $\mathcal{D}$ into small subsets $B_k$, $k = 1, .., N/S_{MB}$ of size $S_{MB} < N$ known as "mini-batches". Then in each step, the gradient is approximated using a single randomly selected mini-batch $B_k$:

$$\nabla_\theta E(\theta) = \sum_{i=1}^{N} \nabla_\theta e(x_i; \theta) \approx \sum_{i \in B_k} \nabla_\theta e(x_i; \theta) = \nabla_\theta^{MB} E(\theta). \tag{1.3}$$

This variant of the GD algorithm is called the stochastic gradient-descent algorithm (SGD). The noise introduced by the sampling of the mini-batches has been linked to the thermal noise in a Langevin trajectory where $\Delta T \sim N/S_{MB}$.

In practice, SGD is often used with a "momentum" term to speed up convergence when close to saddle points (which have zero gradients) and to smoothen sudden variations of the loss landscape. To add this memory term, we modify $v_t$ in Algorithm 1.1.1 to

$$v_t = \gamma v_{t-1} + \eta \cdot \nabla_\theta E(\theta), \tag{1.4}$$

where $0 \leq \gamma < 1$ is known as the momentum hyperparameter or momentum value. SGD with momentum helps the algorithm gain speed in directions with persistent but small gradients, even in the presence of stochasticity, while reducing oscillations in high-curvature directions. An enhanced version of the classical momentum algorithm called Nesterov Accelerated Gradient (NAG) can further amplify these benefits. In NAG the gradient is computed at the expected value of the parameters given our current momentum.

The previously mentioned methods are first-order optimization algorithms that rely on gradients for updating parameters. However, there are higher-order variants that incorporate additional information, such as the Hessian matrix or curvature of the loss function. These methods can provide more accurate and efficient updates, but they can be computationally expensive [9]. A popular alternative is ADAM (Adaptive Moment Estimation),

**Algorithm 2** Stochastic Gradient-Descent with Nesterov momentum (NAG) algorithm

**Require:** $\theta_0$: Initial parameters
**Require:** $\eta$ : Learning rate
**Require:** $\gamma$ : Momentum parameter
**Require:** $E(\theta)$ : Cost function
**while** $\theta_t$ do not converge **do**
Sample a mini-batch (MB) of size $S_{MB}$ from the data set $\mathcal{D}$
$v_t \leftarrow \gamma \cdot v_{t-1} + \eta \cdot \nabla_\theta^{MB} E(\theta_t + \gamma \cdot v_{t-1})$
$\theta_{t+1} \leftarrow \theta_t - v_t$
**end while**

which combines the advantages of adaptive learning rates and momentum-like techniques. While technically a first-order method, ADAM exhibits some benefits of second-order methods, making it a favorable choice for optimization tasks [10]. However, it has been demonstrated that ADAM does not converge to the optimal solution in certain cases and introduces two additional hyperparameters that need to be tuned empirically [11].

We have discussed briefly the basics of ML. We have seen that in order to find an algorithm that indicates how from an input $x$ an output $y$ can be obtained, we need a data set, a model, a cost function, and a learning algorithm. In this report, the models that we use are variational ansatz. Therefore, our focus now shifts toward exploring variational methods and their importance in the study of many-body quantum systems.

## 1.2 Variational methods for wave functions

Variational methods for wave functions are widely used in quantum mechanics to approximate the state of a quantum system. By parameterizing a trial function with adjustable parameters, these methods aim to find the optimal set of parameters that better describe the target state. In the second part of this introduction, we discuss the importance and explore some of the commonly used variational ansatzes to represent quantum states. Lastly, we show how to efficiently compute the expectational values of physical observables for variational states.

### 1.2.1 Variational ansatzes

Storing the quantum state $|\Psi\rangle$ of a quantum system composed of N components involves expanding it over a chosen computational basis $\{|0\rangle, |1\rangle\}^{\otimes N}$. However, this leads to an exponentially increasing array, imposing significant storage limitations due to the exponential growth of memory requirements with system size.

To address these challenges, variational methods offer a promising approach by employing parameterized functions to approximate the wave function called ansatz. The main advantage of variational ansatz lies in their capability to optimally represent complex quantum states using less number of parameters than the components on a computational basis. By exploiting the system's structure and symmetries, variational ansatz efficiently captures the essential properties while minimizing computational resources.

Over the years, various ansatzes have been developed to describe many-body quantum systems. These functions are often inspired by physical phenomena like spin-spin interactions. Here, we explore some of the most commonly used ansatzes for studying many-body quantum systems.

**Mean-field ansatz:** The mean-field approximation assumes that the interaction between N particles can be approximated by the average potential generated by the other particles. Consequently, the wave function can be approximated as the product of single-particle states:

$$|\psi\rangle \approx |\phi_1\rangle \cdot \ldots \cdot |\phi_N\rangle. \tag{1.5}$$

Projecting this function onto an arbitrary element of the Hilbert space basis yields to

$$\psi_\theta(\sigma_1, \ldots, \sigma_N) = \theta_{\sigma_1} \cdot \ldots \cdot \theta_{\sigma_N}, \tag{1.6}$$

where $\sigma_i = \pm 1$. By assuming that $\theta_{\sigma_i} \in \mathbb{R}$, we obtain a variational function with 2N parameters. This mean-field approach allows us to capture the collective behavior of the system. It significantly reduces the computational complexity associated with studying large-scale quantum systems but neglects all quantum effects among degrees of freedom.

**Jastrow ansatz:** The Jastrow ansatz is formulated as a product of two-body correlation functions that depend on the relative positions of pairs of particles within the system. Generally, the Jastrow ansatz can be expressed as

$$\psi_\theta(s_1, \ldots, s_N) = \exp\left(-\sum_{i,j} \theta_{ij}\sigma_i\sigma_j\right). \tag{1.7}$$

This ansatz involves $O(N^2)$ parameters. Moreover, in translationally invariant systems, the parameters $\theta_{ij}$ can be made depending exclusively on the distance between i and j, resulting in $O(N)$ number of parameters.

The Jastrow ansatz allows us to capture the pairwise interactions along a specific axis between particles. However, it neglects the higher-order correlation effects that involve interactions among three or more particles. These higher-order correlations can play a crucial role in certain physical phenomena, especially in systems with strong quantum entanglement or complex collective behavior.

**Artificial Neural Networks (ANN):** Artificial Neural Networks (ANNs) are nonlinear models which were originally loosely inspired by biological networks. Those are widely used in machine learning. ANNs consist of interconnected elements called neurons, denoted by $i$, which take a vector $x = (x_1, \ldots, x_d)$ as input and produce a scalar output $a_i(x)$. These neurons are organized into layers, with each layer serving as the input for the subsequent layer (Fig. 1.1). The first layer is the input layer, followed by hidden layers, and finally, the output layer. The neuron's behavior can be expressed as a nonlinear

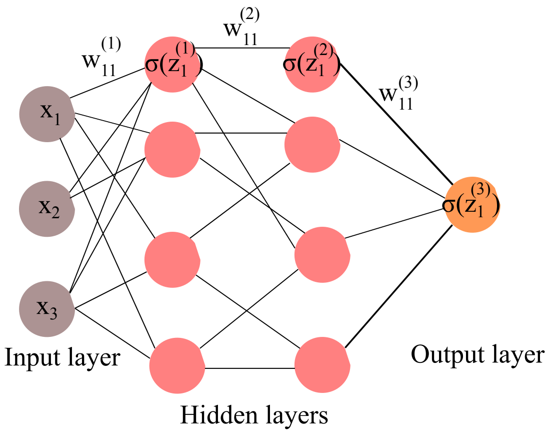


Figure 1.1: Neurons arranged into layers with one layer serving as input to the next layer

function $\sigma$ and a linear function $z$:

$$a_i^{(l)}(z_i^{(l)}) = \sigma\left(\sum_k w_{ik}^{(l)} a_k^{(l-1)} + b_i^{(l)}\right), \tag{1.8}$$

where the superscript $(l)$ indicates the layer and $i$ denotes the neuron within that layer, such that $a_k^{(0)} = x_k$. In Eq. (1.8), the $w_{ik}^{(l)}$ represents the weights of the connection between neuron $k$ and $i$ in layer $l$, while $b_i^{(l)}$ represents the bias associated with neuron $i$. Fig. 1.2 displays common choices of activation functions for real parameters, which introduce nonlinearities into the network. The selection of activation functions influences the computational and training properties of the network, as ANNs are typically trained using gradient-descent-based methods.

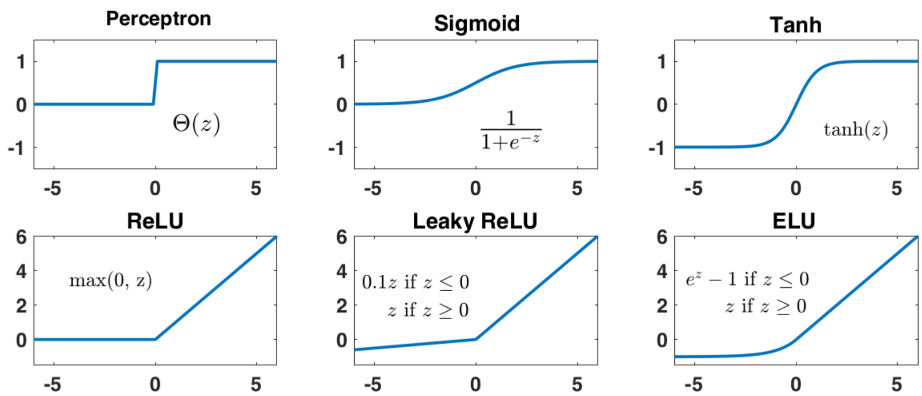


Figure 1.2: Possible nonlinear activation functions for neurons [12]

The effectiveness of ANN is understood in terms of the *Universal Approximation theorem* [13]. Formally, the theorem states that given a function $f : \mathbb{R}^m \to \mathbb{R}^n$ that is continuous on a compact subset of $\mathbb{R}^m$, and a small positive error $\epsilon$, there exists a neural network with a single hidden layer that can approximate $f$ within an error less than $\epsilon$. This approximation can be determined by adjusting the weights and biases of the network. The theorem guarantees the existence of such a network, however, it does not provide a constructive method for finding the exact architecture or parameters. Additionally, the theorem does not specify the number of hidden neurons required to achieve a certain level of accuracy, which can vary depending on the complexity of the function being approximated.

**Restricted Boltzmann Machines (RBMs):** RBMs are an energy-based model that leads to a specific ANN architecture. The energy-based model is described by a Hamiltonian including two sets of neurons: a hidden layer with $N_h$ stochastic neurons, whose values can be either 1 or $-1$, and a visible layer containing $d$ neurons. RBMs are called "restricted" because connections between neurons within the same layer are not allowed. The model is mathematically described by the following expression:

$$\Psi(\boldsymbol{x};\theta) = \sum_{h\in\{-1,1\}^{N_h}} \exp\left(\boldsymbol{h}^\dagger \boldsymbol{\omega} \boldsymbol{x} + \boldsymbol{b}\cdot\boldsymbol{h} + \boldsymbol{a}\cdot\boldsymbol{x}\right), \tag{1.9}$$

where $\boldsymbol{\theta} = \{\boldsymbol{\omega}, \boldsymbol{a}, \boldsymbol{b}\}$ represents a set of parameters. The number of parameters is commonly quantified by the hidden unit density, denoted as $\alpha$, which is defined as the ratio between the number of hidden neurons and the number of particles. For the RBM, the total number of parameters scales as $O(\alpha N^2)$.

By tracing out the hidden neurons analytically, we rewrite the RBM ansatz as

$$\Psi_\theta(x) = e^{\boldsymbol{a}\cdot\boldsymbol{x}} \prod_{i=1}^{N_h} 2\cosh\left(b_i + \boldsymbol{\omega_i}\cdot x\right). \tag{1.10}$$

Thanks to the Universal Approximation Theorem, an RBM can approximate any desired probability distribution with arbitrary accuracy, given a sufficient number of hidden units. This raises the question of whether RBMs can efficiently represent many-body quantum states. To achieve this, it would be necessary to employ complex parameters to represent the phase of the quantum states. Notably, it has been demonstrated that a standard fully connected RBM, where all visible units are connected to each hidden unit, can encode a wave function with volume-law entanglement [14]. The volume law characterizes a situation in which the entanglement entropy scales with the volume of the subsystem, rather than its surface area. This behavior is observed in higher-dimensional systems such as two-dimensional or three-dimensional lattices, where entanglement can spread more uniformly throughout the system.

We have explored several commonly used ansatzes for describing many-body quantum systems. In addition to providing an optimal representation, these ansatzes possess hidden properties that prove to be useful.

### 1.2.2 Expectation Value and Local estimator

Here, we discuss how to efficiently compute expectation values of physical observables for a neural quantum state (NQS) [1]. Further, through the application of statistical sampling

[1] A neural quantum state is a variational state where the ansatz is an artificial neural network.

methods, we show how efficiently estimate the expectation value of an operator.

Let's consider a variational state the ket $|\Psi_\theta\rangle \in \mathcal{H} = Span\left(\{|-1\rangle, |1\rangle\}^{\otimes N}\right)$. We express the expectation value of an operator $\hat{\mathcal{O}}$ in this state as

$$\langle\hat{\mathcal{O}}\rangle = \frac{\langle\Psi_\theta|\,\hat{\mathcal{O}}\,|\Psi_\theta\rangle}{\langle\Psi_\theta|\Psi_\theta\rangle} = \frac{\sum_s |\langle\Psi_\theta|s\rangle|^2 \sum_{s'} \langle s|\,\hat{\mathcal{O}}\,|s'\rangle \frac{\langle s'|\Psi_\theta\rangle}{\langle s|\Psi_\theta\rangle}}{\sum_{s''} |\langle\Psi_\theta|s''\rangle|^2}. \tag{1.11}$$

By defining the Born probability distribution as

$$P(s) = \frac{|\langle\Psi_\theta|s\rangle|^2}{\sum_{s''} |\langle\Psi_\theta|s''\rangle|^2}, \tag{1.12}$$

which indicates the probability that the variational state would be in a certain configuration $|s\rangle \in \{|-1\rangle, |1\rangle\}^{\otimes N}$ if measured along the computational basis. We also introduce the *local estimator* of $\hat{\mathcal{O}}$ as

$$\mathcal{O}_{\text{loc}}(s) = \frac{\langle s|\,\hat{\mathcal{O}}\,|\Psi_\theta\rangle}{\langle s|\Psi_\theta\rangle} = \sum_{s'} O_{ss'} \frac{\Psi_\theta(s')}{\Psi_\theta(s)}, \tag{1.13}$$

we can rewrite the expectation value as

$$\langle\hat{\mathcal{O}}\rangle = \sum_s P(s)\mathcal{O}_{\text{loc}}(s) = \langle\mathcal{O}_{\text{loc}}\rangle_P\,. \tag{1.14}$$

Expressing the expectation value of $\hat{\mathcal{O}}$ as a statistical average allows us to approximate the average with the sample mean, meaning that we can consider a small set of L samples from $P(s)$, denoted as $\{s^{(i)}\}_{i=1}^L$. The sample mean of the expectation value is given by

$$\langle\hat{\mathcal{O}}\rangle \approx \frac{1}{L}\sum_{i=1}^{L} \mathcal{O}_{\text{loc}}(s^{(i)}), \tag{1.15}$$

with the standard error $\epsilon = \sqrt{\frac{\sigma^2}{L}}$, where $\sigma^2 = \langle\hat{O}^2\rangle$. However, generating a set of samples according to the Born probability distribution $\{s^{(i)}\} \sim P(s)$ is generally a non-trivial computational task. In this work, we will employ two sampling methods: Direct Sampling and the Metropolis-Hastings algorithm.

### 1.2.3 Sampling methods

A sampling method, in the context of statistical analysis, is a technique used to obtain samples distributed according to the target distribution. It involves selecting a subset of data points or observations to infer information about the entire dataset.

**Direct Sampling (DS)**: DS refers to a method of generating samples from a probability distribution by directly selecting random points from the distribution without the need for intermediate steps or calculations. It is a straightforward and intuitive approach to sampling, particularly when the probability distribution is known explicitly. The DS algorithm is as follows:

1. Construct the cumulative probability distribution $R(s^{(j)}) = \sum_{i<j} P(s^{(i)})$.

2. Generate a random real number $x \in [0, 1]$ using a probability distribution $Q(x)$ (often uniform distribution is employed).

3. Find $s^{(k)}$ such that $R(s^{(k)}) > Q(x)$.

While the DS offers a reliable approach for generating samples, it is important to consider it has exponential computational complexity. The process of computing and normalizing the probability distribution requires $O(2^N)$ operations and cannot be employed in large systems. As a result, applying this method to high-dimensional probability distributions may not be optimal due to the computational burden involved. Alternative sampling techniques or approximations may be more suitable for efficiently handling large-scale systems and reducing computational costs.

**Metropolis-Hastings algorithm (MH):** MH is a statistical method to generate samples from a (possibly unnormalized) probability distribution. MH to the class of Markov chain Monte Carlo methods (MCMC)[2]. To sample from the target distribution $P(s)$, we follow this procedure:

- Initialize the Markov chain by setting the initial sample $s_0$. This initial sample can be chosen randomly or using a predefined value.

- For each iteration $t$ from 0 to $L$:
  - Propose a new sample $s_{(t+1)}$ according to a transition kernel distribution $T(s_t \to s_{t+1})$.
  - Compute the acceptance probability $A = \min\left(1, \dfrac{P(s_{t+1})T(s_{t+1} \to s_t)}{P(s_t)T(s_t \to s_{t+1})}\right)$, which determines whether to accept or reject the proposed sample.
  - Generate a random number $x$ from a uniform distribution between 0 and 1.
  - If $x < A$, accept the proposed sample $s_{(t+1)}$ and save it as part of the Markov chain. Otherwise, reject it and save the current sample $s_{(t)}$ again.

In this work, we use kernel probability distribution *local flip*. This is a symmetric process, therefore the acceptance probability is reduced to

$$A = \min\left(1, \frac{P(s_{t+1})}{P(s_t)}\right). \tag{1.16}$$

In the MH algorithm, apart from the commonly used local flip, other options for the kernel distribution include Gaussian proposals, multimodal distributions, adaptive proposals, and non-local proposals [15]. Each of these alternatives provides different ways to generate proposals for the next state, allowing for various types of transitions in the state space and enhancing the exploration of the target distribution. The choice of kernel distribution depends on the specific characteristics of the target distribution and the objectives of the sampling process.

[2]A Markov chain is a stochastical model where each event depends only on the state of the previous event. The probability that the next event is $x'^{(t)}$ given the current state $x^{(t)}$ is defined through the kernel distribution, $T(x \to x')$

In this section, we have elucidated the fundamental concepts and methodologies employed in this study. We have demonstrated that when the underlying function generating a dataset is unknown, it is possible to propose a parameterized function capable of describing it. The process of determining the optimal parameters, known as learning, can be achieved through the application of gradient descent methods. In the realm of quantum physics, a similar approach is employed to ascertain the parameters of a variational ansatz that approximates a many-body quantum state. Among the extensively studied ansatzes, the RBM stands out due to its exceptional expressiveness, efficiently representing a quantum system of dimension $2^N$ using $O(\alpha N^2)$ parameters. Another notable advantage of utilizing variational ansatzes is their capacity to estimate the expected value of an operator within a variational state by leveraging local estimators and sampling techniques.

# 2 Variational Monte Carlo of Transverse Field Ising Model

In this section, we provide an overview of the Variational Monte Carlo (VMC) method, a computational approach for approximating the ground state of quantum systems. We demonstrate the application of VMC in finding the ground state of the 1D Transverse Field Ising model Hamiltonian, which serves as a valuable toy model for our study. Additionally, we highlight the significance of this spin model in capturing some aspects of many-body systems. Finally, we present the results obtained from our VMC algorithm, which leverages the ansatz discussed in the previous section, to accurately determine the ground state of the Hamiltonian.

## 2.1 Variational Monte Carlo method

The Variational Monte Carlo method is a computational technique used to approximate the ground state of a quantum system. It combines the principles of Monte Carlo sampling and variational optimization.

Given a Hamiltonian $\hat{H}$, and a variational state $|\Psi_\theta\rangle$, the expectation value of $\hat{H}$ in this state is

$$\langle \hat{H}_\theta \rangle = \frac{\langle \Psi_\theta | \hat{H} | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle}, \tag{2.1}$$

which is bounded from below by $E_0$, the ground state energy of the system represented by $|\Psi_0\rangle$. This implies that by minimizing the expected energy of the variational state, we can determine a set of parameters $\hat{\theta}$ such that the resulting state $|\Psi_{\hat{\theta}}\rangle$ approximates the ground state $|\Psi_0\rangle$. The accuracy of the parametrized state in representing the ground state depends on the specific model and the training algorithm employed.

In order to find the parameters that minimize the expected energy of a Hamiltonian in a variational state, we optimization algorithms such as SGD or ADAM. We define as loss function the expected energy, $\langle \hat{H}_\theta \rangle = E_\theta$, such that its gradient in the parameter space is

$$\nabla_{\theta_k} E_\theta = \sum_s P_\theta(s)(H_{loc}(s) - E_\theta)(D_k(s) + D_k^\dagger(s)), \tag{2.2}$$

where $P_\theta(s)$ is the Born distribution of the variational state, $H_{loc}$ is the local estimator of the Hamiltonian, and $D_k(s)$ is

$$D_k(s) = \nabla_{\theta_k} \log \Psi_\theta(s) = \frac{\langle s | \nabla_{\theta_k} \Psi_\theta \rangle}{\langle s | \Psi_\theta \rangle}. \tag{2.3}$$

Because performing the full sum over the Hilbert space ($\sum_s$) would be exponentially expensive in the system size, we approximate the expression of the gradient with the sample mean,

$$\nabla_{\theta_k} E_\theta \approx \frac{1}{L} \sum_{i=1}^{L} (H_{loc}(s^{(i)}) - E_\theta)(D_k(s^{(i)}) + D_k^\dagger(s^{(i)})), \tag{2.4}$$

where $L$ is the number of samples obtained by sampling from $|\langle s|\Psi_\theta\rangle|^2$.

## 2.2 Transverse Field Ising Model

The Transverse Field Ising model (TFIM) is a paradigmatic quantum mechanical model used to describe the physics of some magnetic materials. It is also known as the Quantum Ising Model, as the transverse field gives off-diagonal terms in the Hamiltonian that cannot be described classically. In the transverse field Ising model, an additional term is introduced, known as the transverse field, which represents an external magnetic field acting perpendicular to the spins. This term can induce quantum fluctuations and lead to interesting quantum phenomena in higher dimensions [16]. We remark that in 1-Dimension it is possible to remap the partition function of the 1D TFIM model onto the partition function of a 2D classical Ising model, meaning that the model is essentially classical at low dimension.
The TFIM Hamiltonian is given as

$$\hat{H}_{TFIM} = -J\left(\sum_{\langle i,j\rangle}\hat{\sigma_i^z}\hat{\sigma_j^z} + g\sum_k \hat{\sigma_k^x}\right), \tag{2.5}$$

where the operators $\hat{\sigma}_i^{z,x}$ are the Pauli matrices at lattice site $i$. In the first term, $J$ is the coupling factor that indicates the strength of the interaction between the adjacent spins. The second term is the transverse field, where $gJ$ is the intensity of the field. The nearest neighbor interaction favors spin alignment ($\uparrow$ or $\downarrow$) while the interaction with the external field favors the alignment in the x-positive direction ($\rightarrow$).

Throughout this work, we consider the values $g = 1$ for the TFIM. This particular choice corresponds to a quantum critical point that separates two distinct phases of the system: the paramagnetic and ferromagnetic phases. Near the critical point, the ground state of the system can exhibit long-range correlations and entanglement, which present difficulties in accurately describing it with a limited parameterized ansatz. Therefore, accurately characterizing the ground state becomes more challenging in this region due to the intricate nature of its correlations and entanglement.

## 2.3 Extracting the ground state from TFIM

Finding the ground state of the TFIM Hamiltonian for systems with a large number of spins using analytical methods can be challenging and computationally expensive. This is due to the exponential increase in the dimension of the Hilbert space, which scales as $2^N$. Consequently, diagonalizing the Hamiltonian matrix requires significant computational resources. Alternatively, VMC methods offer a computationally efficient approach.

We employ the MH sampling method in combination with the SGD algorithm to approximate the ground state wave function of the TFIM Hamiltonian for a system of 10 spins. In Fig.2.1, we present a comparison of the convergence speed among different variational ansatzes discussed in Section 1.2.1 to find the many-body ground state. Notably,

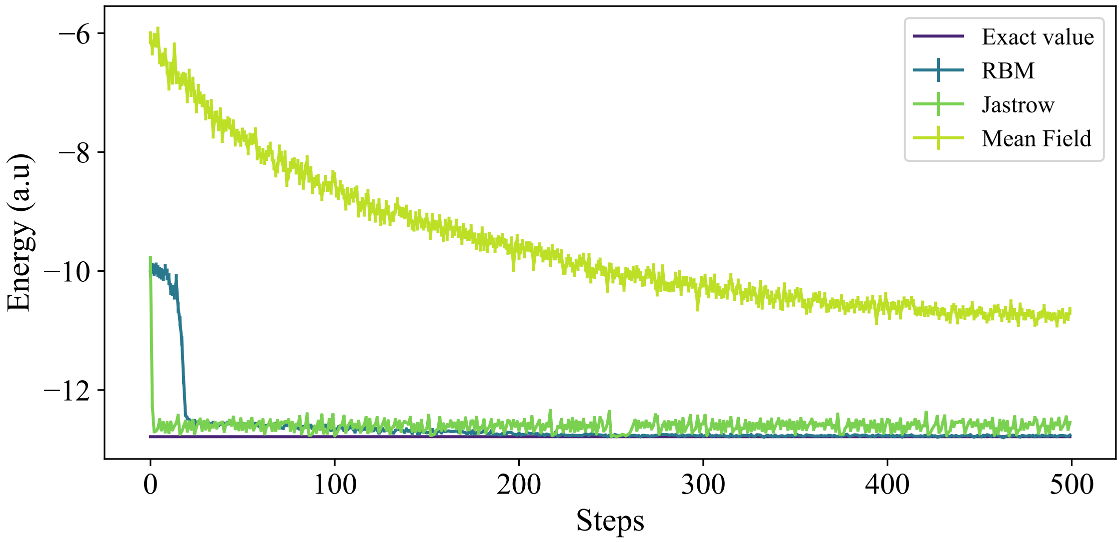


Figure 2.1: **Learning curve of the many-body ground-state energy using different ansatzes**. Expected energy of the variational state $|\Psi_{\theta(t)}\rangle$ (RBM, Jastrow, and Mean Field) at step $t \in [0, 500]$. We estimate the gradient of the energy using 1000 samples from MH algorithm. As a learning algorithm, we use standard SGD(learning rate = 0.05). Besides the exact value is computed using exact diagonalization.

the RBM ansatz, known for its high expressiveness[1], exhibits faster convergence and lower variance compared to other variational wave functions, indicating its superior performance in approximating the desired ground state.

We have provided an overview of the VMC method and applied it to determine the ground state of the TFIM at the quantum critical point. Through the utilization of ML methods, we employed the expected energy of the variational ansatz (RMB, MF, and Jastrow) as the cost function to optimize the parameters that best represent the ground state. Our analysis revealed that among the various ansatzes considered in this study, RBM exhibited superior characteristics in terms of expressiveness, entanglement representation, and efficient variational optimization. Based on these findings, we have chosen to exclusively employ RBM as the ansatz in subsequent sections. This selection ensures a more accurate approximation of the ground state and enhances the overall effectiveness of our approach.

[1]Expressiveness in machine learning refers to the ability of a model or algorithm to capture and represent complex patterns, relationships, and functions within a given dataset.

# 3 Quantum State Tomography

Quantum state tomography (QST) is a technique used to reconstruct a classical copy of the quantum state in a device, such as a quantum computer. It uses repeated measurements on an ensemble of identical quantum states and using the obtained measurement data to infer information about the underlying quantum state. The quantum states can be prepared by various devices or systems, ensuring consistency in the preparation of pure states or general mixed states. By carefully selecting measurement settings and performing a series of measurements, quantum tomography allows us to determine the density matrix or wave function that describes the quantum state. This reconstruction process plays a crucial role in understanding and characterizing quantum systems, enabling us to gain insights into their properties and behavior.

In this chapter, our focus is on reconstructing pure quantum states from numerically generated data, known as synthetic data. In order to achieve this, we show how using ML algorithms and the Kullback-Leibler divergence, we can find the optimal parameters of a NQS whose probability distribution approximates the measurement data. Additionally, we demonstrate the effectiveness of our algorithm by presenting an example of reconstructing the Bell states.

## 3.1 From measurement data to wave function

Kullback-Leibler (KL) divergence, also known as relative entropy, is a measure of the difference between two probability distributions. Specifically, given two probability distributions, $P(x)$ and $Q(x)$, defined over the same discrete sample space $\mathcal{D} = \{x\}$, the KL divergence between them is calculated as

$$KL(P, Q) = \sum_{x \in \mathcal{D}} P(x) \log\left(\frac{P(x)}{Q(x)}\right). \tag{3.1}$$

It is clear that $KL(P, Q) = 0 \Leftrightarrow P(x) = Q(x), \forall x \in \mathcal{D}$. However, KL is not considered a metric because it is not symmetric as $KL(P, Q) \neq KL(Q, P)$. We leverage the concept of KL divergence to perform a comparative analysis of two quantum states.

We consider a quantum state $|\Psi\rangle$, and we perform the measurement of an observable $\hat{A}$. The results will be one of the various eigenvalues of $\hat{A}$, $a_i$, each with probability given by the overlap between the eigenstate $|a_i\rangle$ and $|\Psi\rangle$. The probability distribution of the measurements is therefore $P(a_i) = |\langle a_i|\Psi\rangle|^2 / \langle\Psi|\Psi\rangle$. In order to construct a classical copy of the state $\psi$, we will take the KL divergence between a NQS $|\Psi_\theta\rangle$ and the target distribution $P(a_i)$.

Our classical reconstruction will be encoded into an NQS $|\Psi_\theta\rangle$, for which we may choose any Neural Network architecture but without loss of generality, we will consider here an RBM. The loss function to be minimized is the KL divergence,

$$KL(\theta) = \sum_{a_i} P(a_i) \left[\log\left(\frac{P(a_i)}{\langle a_i|\Psi_\theta\rangle \langle\Psi_\theta|a_i\rangle}\right) + \log\left(\langle\Psi_\theta|\Psi_\theta\rangle\right)\right], \tag{3.2}$$

which expresses the distance between the classical copy (the NQS) and the empirical distribution. In order to use one of the learning algorithms mentioned previously to find the parameters that minimize the KL divergence, we find that[1]

$$\nabla_{\theta_k} KL(\theta) = \langle D_k \rangle_{|\Psi_\theta|^2} - \sum_{a_i} P(a_i) \left[ D_k(a_i) + D_k^\dagger(a_i) \right]. \tag{3.3}$$

If the parameters of the variational state are real: $D_k(a_i) = D_k^\dagger(a_i)$. On the other hand, if the variational ansatz has complex parameters and is holomorphic[2]: $D_k^\dagger(a_i) = 0$.

However, the information about the phase of the system is not encoded in a simple probability distribution. A probability distribution can always be defined from a wave function, but not the reverse. In order to solve this problem, we need to consider measurements performed in different bases associated with a set of observables $\{\hat{A}_j, j = 1, ..., N_B\}$, where each of them has an associated eigenbasis $B_j = \{|a_i^j\rangle, i = 1, ..., 2^N\}$. Indeed, the form of the quantum state on a different basis involves the interference between amplitudes in different bases. Hence, matching the probability distribution defined by snapshots in different measurement bases leads to the correct quantum state as long as the bases contain enough information about the quantum state [17]. Mathematically this implies setting that

$$KL(\theta) = \sum_{j=1}^{N_B} \sum_i P(a_i^j) \left[ \log \left( \frac{P(a_i^j)}{\langle a_i^j | \Psi_\theta \rangle \langle \Psi_\theta | a_i^j \rangle} \right) + \log \left( \langle \Psi_\theta | \Psi_\theta \rangle \right) \right]. \tag{3.4}$$

Calling $\hat{R}^{(0,j)}$ to the unitary operator that transforms the the basis $B_j$ into $B_0$, such that $\hat{R}^{(0,j)} |a_i^j\rangle = |a_i^0\rangle$. We rewrite the previous equation as

$$KL(\theta) = N_B \log \left( \langle \Psi_\theta | \Psi_\theta \rangle \right) + \sum_{j,i} P(a_i^j) \log \left( \frac{P(a_i^j)}{| \langle a_i^0 | \hat{R}^{(0,j)\dagger} | \Psi_\theta \rangle |^2} \right). \tag{3.5}$$

Then, we taking the gradient on the parameter space, we get

$$\nabla_{\theta_k} KL(\theta) = N_B \langle D_k \rangle_{|\Psi_\theta|^2} + \sum_{j,i} P(a_i^j) \left[ \langle D_k(a_l^0) \rangle_{Q_{ij}} + \langle D_k^\dagger(a_l^0) \rangle_{Q_{ij}^*} \right], \tag{3.6}$$

where $Q_{ij}$ is a pseudo-probability distribution given as

$$Q_{ij}(a_l^0) = R_{il}^{(0,j)\dagger} \Psi_\theta(a_l^0), \tag{3.7}$$

and $R_{ik}^{(0,j)\dagger}$ is the matrix representation of the operator $\hat{R}^{(0,j)\dagger}$ in the basis $\{|a_i^0\rangle\}$.

Before concluding this section, it is important to note that when working with a variational ansatz that involves complex parameters, the update rule becomes [18]:

$$\theta_{t+1} \leftarrow \theta_t - \kappa \left[ \nabla_\theta \mathcal{L}(\theta_t) \right]^*, \tag{3.8}$$

where $\kappa$ is a learning rate. Throughout the remainder of this work, we refer to the ML algorithm that employs the KL divergence as the cost function and RBM as ansatz as the

[1]The notation $\langle F \rangle_G$ means the average of the function $F$ on the probability distribution $G$, $\langle F \rangle_G = \sum_x G(x)F(x)/\sum_x G(x)$

[2]If a function $f(z)$ is holomorphic then $\partial f/\partial \bar{z} = 0$. RBM with complex parameters is holomorphic.

*Reconstruction algorithm.*

## 3.2 Preliminary results, Bell states

We develop a reconstruction algorithm which takes data of observable measurements and then uses the KL divergence as a loss function in a stochastic gradient descent algorithm to find the complex parameters of an RBM ansatz that approximates the wave function associated with the measurements. In order to test this algorithm, we reconstruct the wavefunctions of the Bell states from synthetic data. Furthermore, we measure the distance between $|\Psi_\theta\rangle$ and the target state $|\phi\rangle$ by computing the infidelity, which is defined as:

$$Inf\left(|\Psi_\theta\rangle, |\phi\rangle\right) = 1 - \left|\sum_\sigma \langle\phi|\sigma\rangle \langle\sigma|\Psi_\theta\rangle\right|^2. \tag{3.9}$$

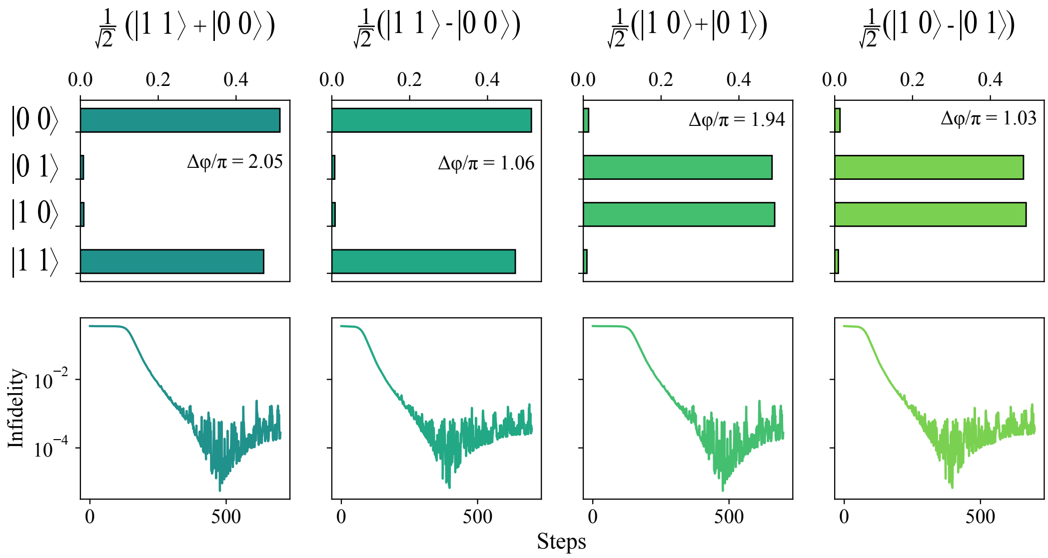


Figure 3.1: **Preliminary reconstructions of the Bell states from synthetic data.** Here we display the probability distribution of the reconstructed Bell states and phase differences between the main components. Also, the Infidelity between the reconstructed state and the target state is computed at each iteration of the learning process. The training is done using RBM with a hidden unit density of 1 as an ansatz. As an optimizer, we use Vanilla SG with a learning rate of 0.03 and 700 iterations.

In Fig. 3.1, we present the reconstruction of the Bell states using a synthetic dataset of size 1000. The dataset was generated by measuring on the bases $ZZ$, $XZ$, $XX$, $ZX$, $YY$, $YZ$, and $ZY$. The probability distributions of the reconstructed states in the figure align with the expected ones, and the phase difference between the main components is close to the expected values. Additionally, we observe a decrease in the Infidelity with the number of iterations, indicating that the variational states are approaching the synthetic state. However, the quality of the reconstruction could be improved. Since this is a preliminary result, the hyperparameters of the gradient-descent optimizer were not tuned correctly.

Nevertheless, this initial result confirms the functionality of the algorithm, and further improvements can be achieved by utilizing more advanced optimizers such as ADAM or NAG and increasing the number of iterations.

We have presented an ML algorithm for reconstructing the wave function based on synthetic measurement data. The KL divergence served as a measure of similarity between the synthetic and variational probability distributions. By employing this divergence as the cost function in an ML algorithm, we obtained the optimal parameters of the variational ansatz, providing an approximation of the unknown quantum system underlying the measurement data. To validate the algorithm, we applied it to reconstruct the wave function of the Bell states, demonstrating that the obtained amplitudes and phases corresponded to the target states. To quantify the "distance" between the target and variational states, we computed the *Infidelity*, which we utilize throughout this work. In the subsequent section, we leverage this algorithm to reconstruct the ground state of the TFIM Hamiltonian.

# 4 Reconstructing ground state of TFIM model from data

In this section, we evaluate the performance of the reconstruction algorithm using synthetic data obtained from the ground state of the 1D TFIM. We begin by examining the training process and introducing the metrics employed to assess the reconstruction quality. Subsequently, we address the challenges that can hinder achieving optimal convergence in the reconstruction process. To overcome these challenges, we propose an enhanced method, called *Control variates*, designed to improve the quality of the reconstructions and enable closer convergence to global minima. Additionally, we investigate the scalability of the algorithm's accuracy with respect to the dataset size. Lastly, we assess the algorithm's robustness by testing it on a more realistic synthetic measurement dataset sourced by quantum noise models.

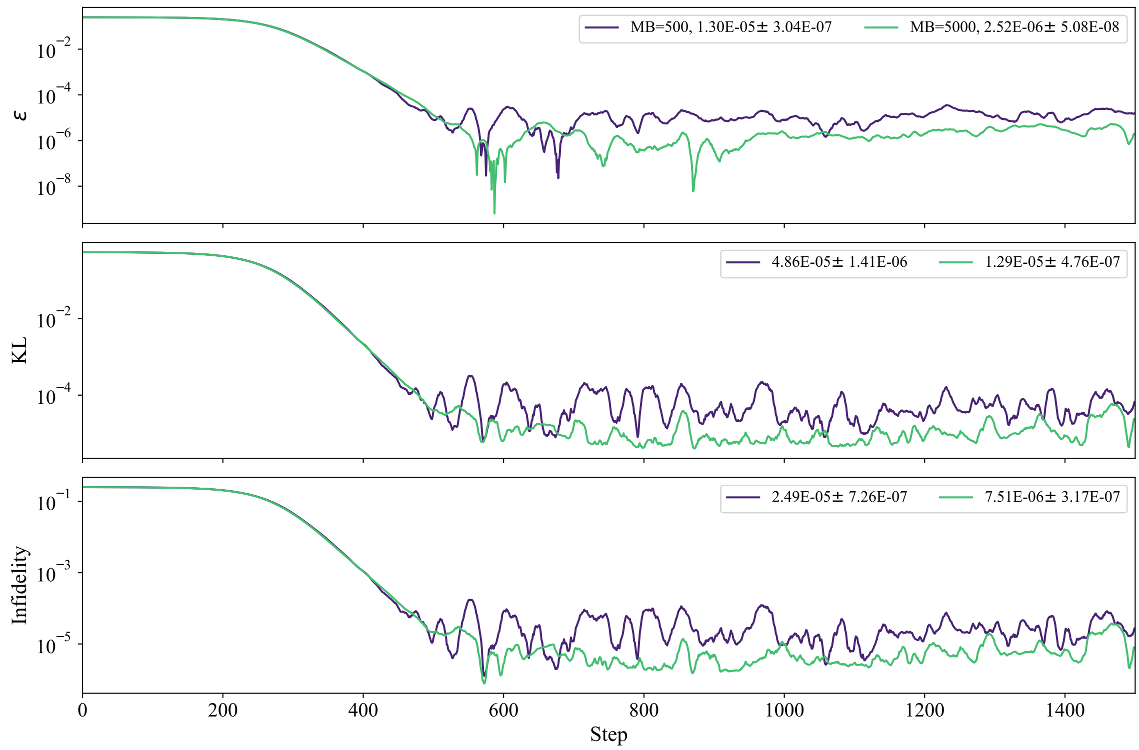


Figure 4.1: **Learning curves for TFIM with 3 spins.** This figure illustrates the learning process using the RBM ansatz with hidden neurons density of $\alpha = 3$ and real parameters for the TFIM problem. We examine the effect of two different mini-batch sizes (MB) on the learning performance, considering a data size of $N_S = 10^6$. Once convergence is achieved, we calculate the average and variance of the last 500 data points for evaluation. The relative energy error ($\epsilon$) and KL divergence serve as the evaluation metrics for performance analysis. The training procedure employs the NAG algorithm with a learning rate of $10^{-3}$ and Nesterov momentum set to 0.9. Unless mentioned, the same procedure is done in all the following reconstructions.

## 4.1 Effect of minibatching on the reconstruction accuracy

We employ the Neural Quantum State Reconstruction algorithm discussed previously to perform a tomographic reconstruction of synthetic data representing the ground state of a 3-spin TFIM. We obtain the ground state through some exact technique and we generate from it the dataset by sampling. Because the Hamiltonian is stoquastic[1], our focus is solely on the $Z$ basis, which proves sufficient for studying the ground state of the TFIM.

In Fig.4.1, we present the learning curves for the relative energy error ($\epsilon$), KL divergence, and infidelity. It is evident that the error $\epsilon$ decreases as the number of iterations increases, indicating the convergence of the expected energy of the variational state towards the ground state energy of the many-body system. Similarly, both infidelity and KL divergence display a similar trend. However, all three metrics eventually reach a point where they exhibit slow decreases and fluctuations. These fluctuations can be attributed to several factors, including the learning rate and the mini-batch size. The learning rate presents a fundamental challenge in ML algorithms, as a finite learning rate prevents us from reaching the true global minimum of the loss function. While choosing a smaller learning rate can bring us closer to the minimum, it may require more iterations to achieve convergence. Regarding the mini-batch size, as shown in Fig.4.1, increasing it by an order of magnitude reduces the variance of the metrics, resulting in smaller values that indicate more accurate parameter estimation.

In Fig.4.2 we investigate the effect of mini-batch size on the accuracy of the reconstructed state. We observe that a larger mini-batch size leads to better accuracy, suggesting that the noise introduced by smaller batches degrades the quality of the gradient at smaller errors. When the MB size is large, it provides a better representation of the entire data set, minimizing the differences between MBs used during the algorithm. Consequently, the computation of the gradient of the loss function between subsequent iterations becomes more consistent, reducing the noise observed near the convergence region. The robustness of gradient estimates allows for improved precision and smoother adjustments to the model parameters during the training process. However, it is important to note that increasing the MB size indefinitely does not guarantee a continuous improvement in accuracy. In Fig., where a data size of $10^6$ is considered, we observe that the relative error and other metrics initially converge to smaller values as the MB size increases. However, a saturation point is reached when the MB size is around $\approx 10^4$. Further increasing the MB size does not significantly reduce the metrics. Determining the optimal MB size is a challenging task that depends on the specific problem and data set size, and it remains an ongoing topic of study in the machine learning community [19].

[1]Hamiltonians, where all the off-diagonal elements in the standard basis are real and non-positive, are called "stoquastic".

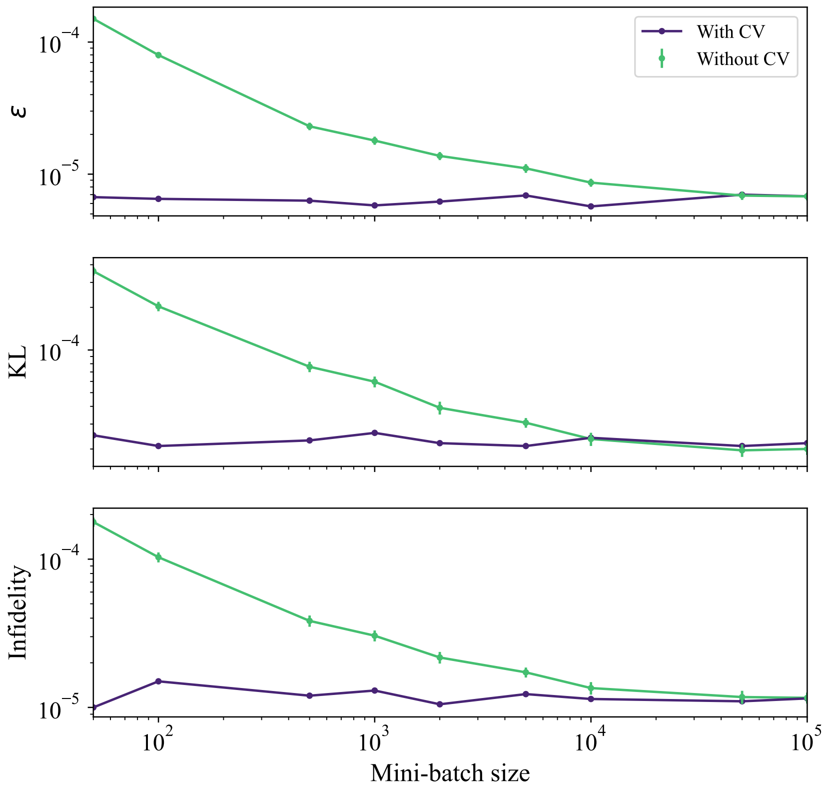


Figure 4.2: **Effect of Mini-Batch Size on KL Divergence, Relative Energy Error, and Infidelity: With and Without Control Variate.** We show how the quality of the reconstruction state depends on the mini-batch size with and without Control Variate (CV) set to $\tau = 30$. The training is done for 3000 iterations using a data set with $N_S = 10^6$ samples.

In the next section, we demonstrate a practical approach to eliminate the necessity for fine-tuning the minibatch size and mitigate its adverse impacts. By leveraging this technique, we effectively address the drawbacks of an inappropriate minibatch size, resulting in enhanced model performance and training efficiency.

## 4.2 Control variates

Control variates (CV) is a widely used technique in statistical estimation and machine learning that aims to reduce variance and improve the accuracy of stochastic estimates. It involves introducing an auxiliary variable or function that is correlated with the target variable of interest. Incorporating this additional variable into the estimation process helps to capture and account for some of the variations in the target variable, thus reducing the noise and improving the precision of the estimates.

Computing

$$A = \mathbb{E}(V)_P \tag{4.1}$$

could require a lot of computational resources. By sampling from $P(x)$, we can approximate $A$ to

$$A \approx \frac{1}{L} \sum_{x_s \in D} V(x_s), \quad D = \{x_1, , , x_L\} \tag{4.2}$$

with an error $\epsilon \approx \frac{\sigma_V}{\sqrt{L}}$.

To reduce the error we need to reduce the variance.
Defining:

$$Z(x) = V(x) - \alpha(W(x) - \mathbb{E}(W)), \tag{4.3}$$

$\mathbb{E}(W)$ is known. We see that $\mathbb{E}(Z) = \mathbb{E}(V) = A$. But

$$Var(Z) = Var(V) - 2\alpha Cov(W, V) + \alpha^2 Var(W), \tag{4.4}$$

where the value of $\alpha$ that minimizes (4.4) is $\alpha^* = Cov(V, W)/Var(W)$, such that

$$Var(Z) = Var(V)(1 - Corr(V, W)^2). \tag{4.5}$$

Thus to reduce $Var(Z)$, $W(x)$ and $V(x)$ need to be highly correlated.

In our work, $V \equiv D_k(a_i; \theta_t)$ and $\mathbb{E}(V) \equiv \sum_{a_i} P(a_i)V(a_i)$ [2]. To identify a strongly correlated variable that we can compute, we consider the gradient at a previous iteration. It becomes apparent that the gradient evaluated in the parameters found in the iteration $t-\tau$, denoted as $W \equiv D_k(a_i; \theta_{t-\tau})$, exhibits a strong correlation with $V$. Ideally, setting $\tau$ to 1 would capture the immediate correlation. However, computing the gradient over the entire dataset at each iteration is computationally expensive. Hence, we must choose a value for $\tau$ that balances computational efficiency and correlation strength. A large value of $\tau$ might result in a weak correlation ($\alpha^* \approx 0$), while a small value of $\tau$ may slow down the algorithm. Determining an appropriate value for $\tau$ is crucial for achieving an effective trade-off between computational cost and the desired correlation.

In order to incorporate the control variate (CV) into our developed algorithm, we modify eq.(3.3) as follows:

$$\begin{aligned} \sum_{a_i \in MB} P(a_i) D_k(a_i; \theta(t)) \rightarrow & \sum_{a_i \in MB} P(a_i) D_k(a_i; \theta(t)) \\ & - \alpha^* \left[ D_k(a_i; \theta(t-\tau)) + \sum_{a_j \in \mathcal{D}} P(a_j) D_k(a_j; \theta(t-\tau)) \right]. \end{aligned} \tag{4.6}$$

In Fig.4.2, we observe the impact of the control variate ($\tau = 30$) on the algorithm's performance. Regardless of the mini-batch (MB) size, the evaluation metrics consistently converge to a similar value near the convergence region. Henceforth, we employ the control variate with $\tau = 30$, and the MB size is set to 500 for the remainder of this work.

[2] See eq.(3.3). $D_k(a_i) = \nabla_{\theta_k} \log \langle a_i | \Psi_\theta \rangle$

## 4.3 Scailing of the QST accuracy with the data set size

In the preceding section, we established that the mini-batch (MB) size has a significant influence on the accuracy of quantum state reconstruction. However, we were able to alleviate this dependency by introducing control variates (CV), which effectively reduced the variance resulting from the utilization of different MBs throughout the learning process. With this foundational understanding in place, we now turn our attention to exploring the impact of the number of parameters on the quality of the reconstruction. Additionally, we seek to investigate the scaling behavior of the evaluation metrics in relation to the size of the data set. By delving into these aspects, we aim to gain further insights into the behavior and performance of our reconstruction algorithm.

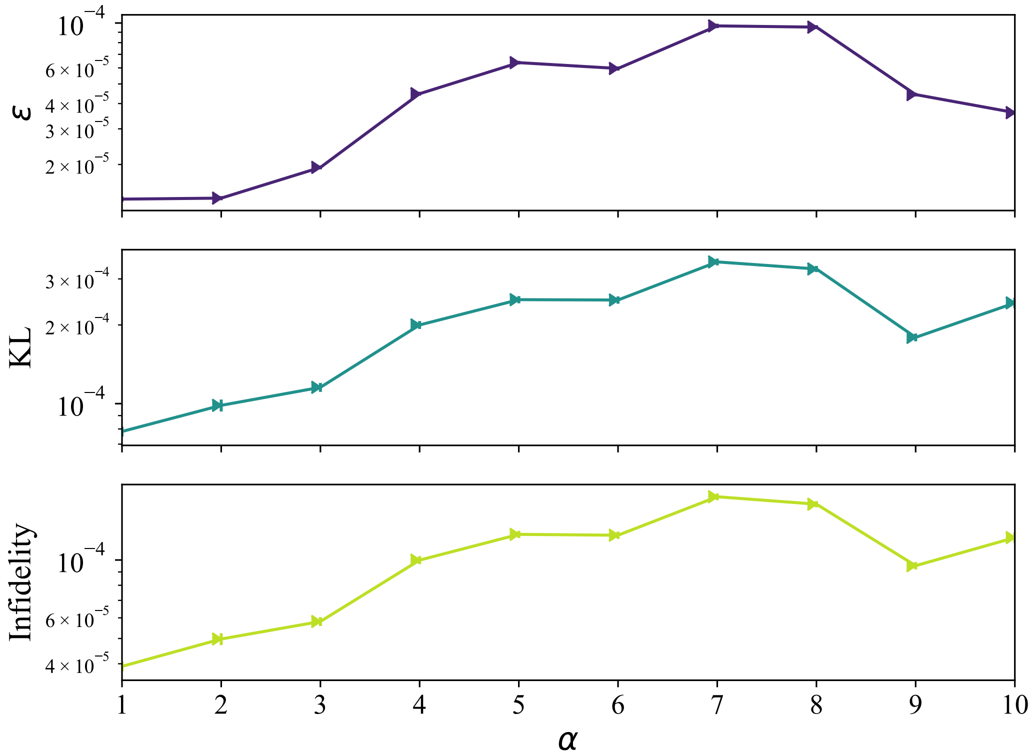


Figure 4.3: **Scaling of Metrics with Hidden Unit Density for TFIM with 4 Spins.** This figure demonstrates the influence of adjusting the hidden unit density ($\alpha$), which relates to the RBM's expressiveness, on the accuracy of the reconstructed state. The training process consists of 20000 iterations.

In Fig.4.3, we present the scaling behavior of the evaluation metrics with respect to the hidden unit density $\alpha$ of the RBM ansatz. It is important to note that $\alpha$ is defined as the ratio of the number of hidden neurons $N_h$ to the total number of spins $N$. Surprisingly, contrary to previous results shown in [20], we observe that increasing the number of hidden neurons does not result in increasing the accuracy of the reconstructed state. This unexpected outcome may be attributed to the fact that training becomes less challenging when there are fewer neurons, as evidenced by the case of $\alpha = 1$. Consequently, to achieve improved results, might be necessary not only to increase the value of $\alpha$, but also to reduce the learning rate, perform more iterations, and optimize the hyperparameters of our learning algorithm, such as the momentum value in the NAG algorithm. However, determining the optimal hyperparameter values remains an empirical endeavor.

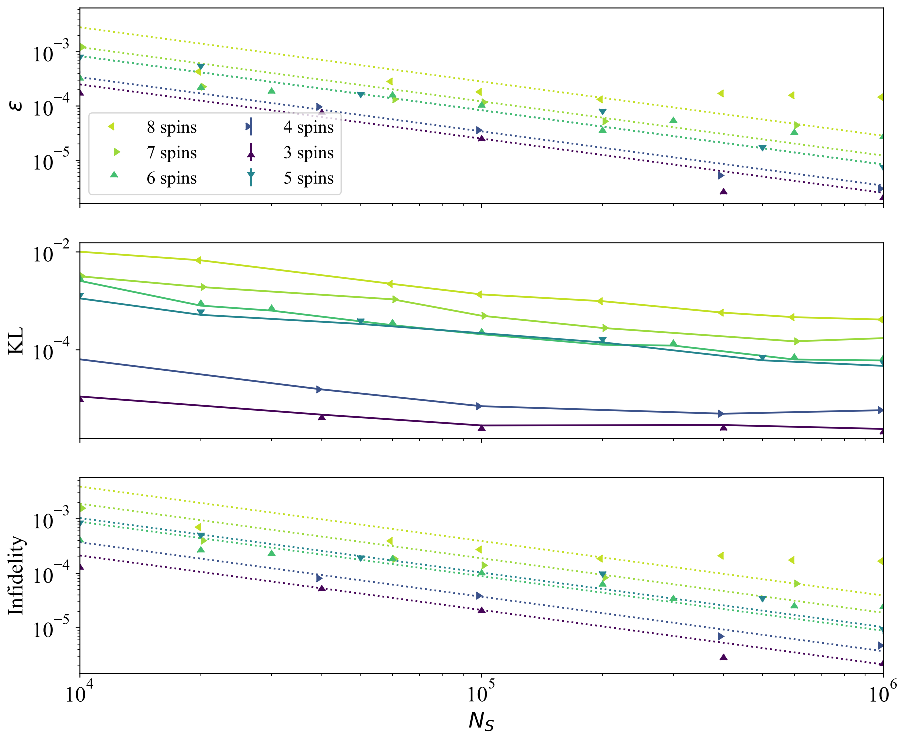


Figure 4.4: **Scaling of the accuracy of the reconstructed state with the dataset Size.** This figure illustrates the behavior of the metrics in the convergence region as the dataset size is varied while keeping the remaining hyperparameters fixed. For configurations involving 3 and 4 spins, we set $\alpha = 4$, and employ the NAG optimizer with a momentum value of 0.3 and a learning rate of 0.003. For configurations with 5 and 6 spins, $\alpha$ is set to 5, and for 7 and 8 spins, $\alpha$ is set to 6. In these configurations, we utilize the NAG optimizer with a momentum value of 0.9, and adopt a learning rate schedule of the form $\eta(1 + t\xi)^{-1}$, where $\eta = 0.01$ and $\xi = 5 \times 10^{-4}$.

In Fig.4.4, we investigate the impact of the dataset size on the accuracy of the reconstructed state. Our analysis reveals that for spin configurations with 3, 4, and 5 spins, a clear power-law scaling behavior is observed, indicating that the coarseness reduces as the dataset size increases following $O(N_s^{-1})$. However, for higher spin configurations, we observe a different scaling pattern that deviates from the expected power law. This suggests the presence of additional factors influencing the accuracy of the reconstructions, potentially related to the optimization parameters. Interestingly, these observations align with the findings presented in [21], where the accuracy of the reconstruction for 8 spins using NQS was shown to scale as $O(N_s^{-1})$. Notably, the study in [21] utilized the ADAM optimizer and a significantly larger number of iterations (around $10^8$) compared to our maximum of $10^5$ iterations. Consequently, it is advisable to scrutinize the optimizer settings to enhance the accuracy of the reconstructions, particularly when dealing with larger datasets of size $N_s > 10^5$.

In Figure 4.5, we investigate the relationship between the prefactor $a$ and the number of spins, revealing the power-law scaling behavior $O(aN_s^{-1})$ that governs the accuracy of the reconstructed state. Our analysis uncovers that both the relative energy error and infidelity metrics exhibit exponential scaling with the number of spins in the convergence region. This observation indicates that for a system with N spins, the accuracy of its reconstruction follows the scaling law $O\left(e^{cN}N_s^{-1}\right)$, where the values of $c_\epsilon$ and $c_{infi}$ are determined to be $0.45 \pm 0.055$ and $0.55 \pm 0.063$, respectively. This result underscores the need for a larger amount of measurement data to accurately reconstruct the quantum state using the NQS method as the system size increases. However, in the case of the model under investigation, we observe a significant reduction in the number of measurement data needed to reconstruct a quantum state with a desired level of accuracy, in contrast to alternative methods where the quality of the reconstructed state scales as $N_s^{-1/2}$ [22].

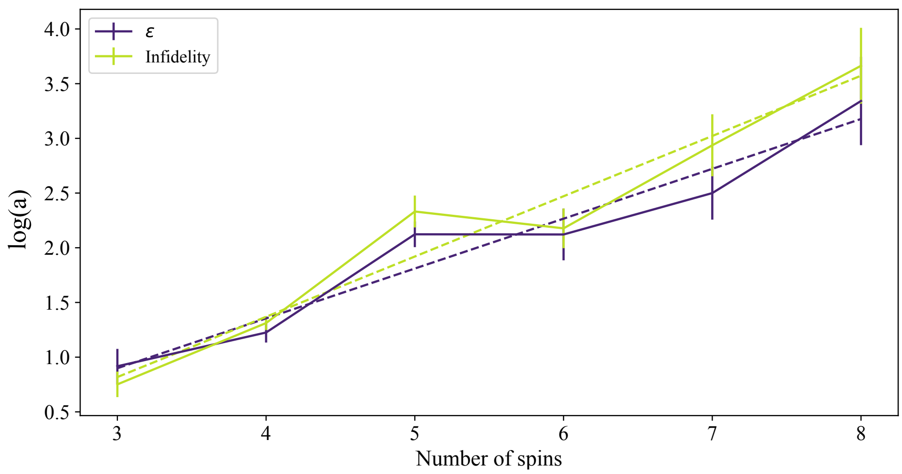


Figure 4.5: **Accuracy scaling with the system's size.** For the relative energy error $\epsilon$ and the Infidelity, we show the dependence on the number of spins of their intercept value $\log(a)$ obtained from the linear regression with the log of the dataset size assuming a fixed slope of -1. For both, we find that $\log(a)$ depends linearly on the system's size. The p-values for $\epsilon$ and infidelity are $1.1E-3$ and $9.4E-4$ respectively.

## 4.4 Reconstructing from non-isolated spins

We evaluate the performance of the reconstruction algorithm using a more realistic dataset obtained by sampling from mixed states. Specifically, we focus on a system consisting of 3 spins that are immersed in a thermal bath. Additionally, we examine another system of 3 spins whose state is characterized by the quantum depolarization noise model.

Considering that the state of a many-body quantum system described by the TFIM immersed in a thermal bath is given by the density operator:

$$\hat{\rho_T} = \frac{e^{-\hat{H}_{TFIM}\beta}}{\mathrm{Tr}\left(e^{-\hat{H}_{TFIM}\beta}\right)}. \tag{4.7}$$

Using sampling methods, we sample from it a dataset of size $10^5$ then, we employ this synthetic data to train the reconstruction algorithm.

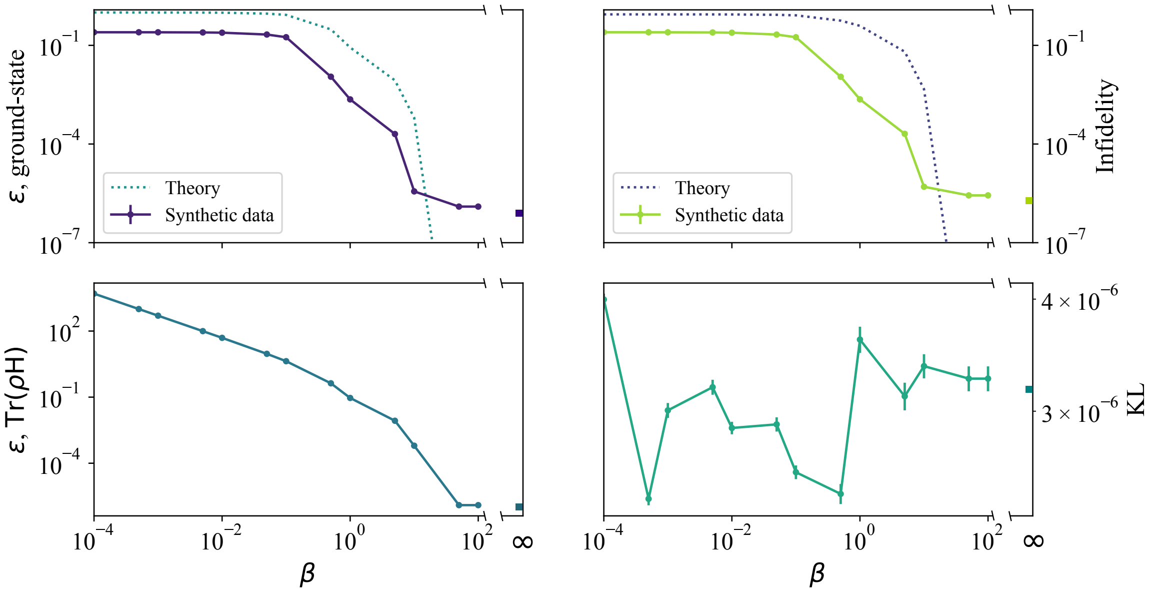


Figure 4.6: **Reconstruction of the State of a TFIM with 3 Spins in a Thermal Bath.** This figure presents the evaluation metrics for the reconstructed state: relative energy error with respect to the ground state and expected ensemble energy, infidelity, and KL divergence. The metrics are plotted as a function of $\beta = J/k_B T$. We generate a dataset comprising $10^6$ samples for various $\beta$ values and utilize them as input for the reconstruction algorithm. The RBM ansatz with $\alpha = 5$ is employed, along with the Nesterov Accelerated Gradient (NAG) learning algorithm using a momentum value of 0.9, a learning rate of $10^{-3}$, and a total of 15,000 iterations.

In Fig.4.6, we show the accuracy of the reconstructed state for different values of $\beta$. We observe that the reconstructed relative energy error with respect to the ground state energy and the infidelity is lower compared to the theoretical values for low values of $\beta$. This indicates that at low $\beta$ values ($\beta < 0.1$), where the mixedness of the state is high, a simple wave function representation is insufficient to accurately capture the measurement data. As $\beta$ increases ($\beta > 0.1$), the mixedness of the objective state decreases and approaches that of a pure state. Consequently, the reconstruction algorithm identifies the parameters that represent this state with a reasonable level of accuracy. Notably, for low temperatures, $\beta \gtrsim 10^2$, the accuracy of the metrics achieves a similar order of magnitude., showing that the quality of the reconstruction may be limited by the dataset size.

On the other hand, we test the algorithm in synthetic data obtained using the quantum depolarization noise model. This model plays a crucial role in the field of quantum information theory, as it captures a fundamental aspect of noise and disturbances that can affect quantum states. In many practical scenarios, quantum systems are subject to environmental interactions and imperfections, leading to the degradation of the originally pure states [23]. The quantum depolarizing noise model provides a mathematical framework to describe such noise processes. It represents the transformation of a pure quantum state, into a maximally mixed state, where the purity of the state is reduced, and a certain level of noise is introduced, mathematically is given as

$$\hat{\rho} = (1-\eta)\left|\Psi_0\right\rangle\left\langle\Psi_0\right| + \eta\frac{\hat{I}}{d}, \tag{4.8}$$

where $d$ is the dimension of the quantum system, and we consider $|\Psi_0\rangle$ as the ground state of the TFIM for 3 spins.

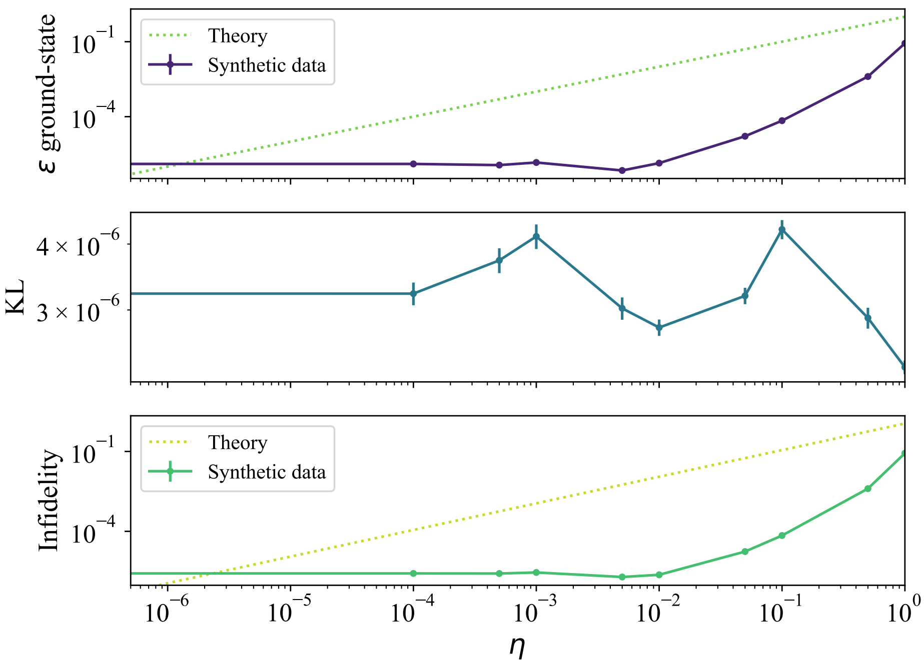


Figure 4.7: **Quality of the reconstructed state from noisy synthetic data.** The theory dotted curve indicates the values that the reconstructed state, solid curve, should take if the model could reproduce the mixedness of the target state. The algorithm is trained using the NAG optimizer with 0.9 momenta and a learning rate of $10^{-}3$ for 15000 iterations

In Fig.4.7, we show the quality of the reconstructed state using synthetic data sampled from 4.8 at different levels of noisiness. We see that the algorithm can reconstruct the ground state with a relatively high level of accuracy for noise levels $\eta < 10^{-2}$. Indicating that the model is stable under small perturbations in the target state. The theoretical curves, which represent the expected metric values if the neural quantum state could fully characterize the system, are consistently higher than the reconstructed values for $\eta > 10^{-6}$. This may show that below this threshold, the accuracy of the model is limited by the dataset size. Overall, for noise levels $\eta < 10^{-2}$, the algorithm demonstrates good performance, as indicated by the similar metric values observed in this entire range.

# 5 Conclusion

In this report, we explore the use of quantum neural states and machine learning algorithms for efficient reconstruction of the ground state of the 1D Transverse Field Ising model. Specifically, we employ Restricted Boltzmann Machines as neural network ansatz due to their high expressiveness, which scales with the hidden unit density. Using synthetic datasets for different sizes of spin systems, we calculate the sample and the system's size complexity of the accuracy of the reconstruction state. Additionally, we reconstruct the quantum state from more realistic synthetic datasets, generated using known noise models, to assess the performance of the ML algorithm.

The obtained results indicate that upon achieving convergence, the relative energy error and infidelity exhibit a scaling behavior with respect to the dataset size, $N_s$, following a decay of $N_s^{-1}$ rather than the $\sqrt{N_s}$ scaling observed in alternative methods. However, it is important to note that for the specific system under investigation of TFIM, the accuracy of the reconstructed state shows also an exponential dependence on the number of system components.

On the other hand, by reconstructing the state from noisy datasets, we find that the algorithm is stable under small perturbations on the target state. Because the model is made to represent a target wave function, it cannot capture the statistical nature of mixed states. As a result, the model interprets the statistical nature of the mixed states as sampling noise, which is overlooked due to the presence of a large number of training samples and the complexity of the model.

In conclusion, current quantum technologies could benefit greatly from machine learning techniques. By approximating a target state to a neural quantum state, we can significantly reduce the exponential scaling of memory required to store the quantum state of a many-body system. Additionally, it allows for a drastic reduction in the number of measurements needed to reconstruct a quantum state with a given accuracy, compared to alternative methods that scale as $N_s^{-1/2}$. The combination of machine learning and quantum technologies presents a promising avenue for overcoming the challenges associated with quantum state reconstruction.

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